THE FINITE ELEMENT METHOD (FEM) ANALYSIS REPORT

Report No: 21-MP-074-02-01

<05/01/2021>

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TABLE OF CONTENTS

1. INTRODUCTION **3**
1.1 Objective 3
2. TEST SUMMARY **6**
3. METHODOLOGY **6**
3.1. Material Information 7
3.2. Geometry of Test models 7
3.3. Loads and Supoorts 10
3.4. Mesh Model of Dental Implants 10
4. TEST RESULTS **11**
5. CONCLUSION **14**
6. TEST SUMMARY REPORT APPROVAL **15**
7. REFERENCES **15**

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1. INTRODUCTION

1.1 OBJECTIVE

The purpose of this test described here is to determine test results of *Dental Implant Torsion Test of Implant Body-Connecting part joints of Endosseous Dental Implants Systems Test FEA Report* in accordance with ISO-TS-13498

In biomechanical analysis, the longest and thinnest implant was selected for worst case selections, and as a result of this selection, the relevant implants and abutments used were marked in the table 1 as seen below. (these products were determined using ref 7.3 -Medisolaris Product List.)

A matrix was created by determining the weakest products in each group. Finite element analysis of the products determined in this matrix was performed and worst case products were determined.

Then the products in table 1 were listed as in table 2.

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TABLE 1. Dental Implants Assembly Matrix

IMPLANT			A1	A2	A3
ABUTMENT			SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 3716- Medisolaris Merkür Hex İmplant Ø3.7 x 16.0 mm Platform Ø3.5 mm
B1	Venüs Con DÜZ (SİMANTE EDİLEBİLİR) ABUTMENT	SLV NP205 Düz Abutment Ø 3.8 mm H:5 mm for Ø3.2 /3.7/4.3/5.0 implant için	X		
B2	Merkür Hex DÜZ (SİMANTE EDİLEBİLİR) ABUTMENT	SLM NP205 Düz Abutment Ø 3.8 mm H:5 mm for Ø3.2 implant için		X	
B3	Venüs Con AÇILI ABUTMENT	SLV NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 /3.7/4.3/5.0 implant için	X		
B4	Merkür Hex AÇILI ABUTMENT	SLM NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 implant için		X	
B5	Venüs Con ANATOMİK DÜZ ABUTMENT	SLV NP303 Düz Anatomik Abutment Ø 3.8 mm H:3 mm for Ø3.2 /3.7/4.3/5.0 implant için	X		
B6	Merkür Hex ANATOMİK DÜZ ABUTMENT	SLM NP303 Anatomik Düz Abutment Ø 3.8 mm H:3 mm for Ø3.2 implant için		X	
B7	Venüs Con ANATOMİK AÇILI ABUTMENT	SLV NP362 Anatomik Açılı Abutment 25° Ø 3.8 mm H:3 mm for Ø3.2 /3.7/4.3/5.0 implant için	X		
B8	Merkür Hex ANATOMİK AÇILI ABUTMENT	SLM NP362 Anatomik Açılı Abutment 25° Ø 3.8 mm H:3 mm for Ø3.2 implant için		X	
B9	Venüs Con MULTİ UNİT ABUTMENT	SLV 30505 Multi Unit Abutment 0° Hex Ø 5.0 mm H:5 mm for Ø3.2 /3.7/4.3/5.0implant için	X		
B10	Merkür Hex MULTİ UNİT ABUTMENT	SLM 30505 Multi Unit Abutment 0° Hex Ø 5.0 mm H:5mm for Ø3.2 implant için		X	
B11	Venüs Con MULTI UNIT AÇILI ABUTMENT	SLV 30516 Multi Unit Açılı Abutment 30° Ø 5.0 mm H:4.5 mm for Ø3.2 /3.7/4.3/5.0 implant için	X		
B12	Merkür Hex MULTI UNIT AÇILI ABUTMENT	SLM 35516 Multi Unit Açılı Abutment 30° Ø 5.0 mm H:4.5 mm for Ø3.7/4.2/4.7/5.2 implant için			X
B13	Venüs Con TOPUZ ABUTMENT	SLV 30605 Topuz Abutment Ø 4,5 mm H:5 mm for Ø3.2 /3.7/4.3/5.0 implant için	X		
B14	Merkür Hex TOPUZ ABUTMENT	SLM 30605 Topuz Abutment Ø 4,5 mm H:5 mm for Ø3.2 implant için		X	
B15	Venüs Con Tibase CAD-CAM ABUTMENT	SLV 30803 Tibase cad-cam Abutment H:3 mm Ø 3.8 mm	X		
B16	Merkür Hex Tibase CAD-CAM ABUTMENT	SLM 30804 Tibase cad-cam Abutment H:3 mm Ø 3.8 mm		X	

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TABLE 2. Dental Implants Assembly List

NO	ABUTMENT	IMPLANT	SCREW
1	SLV NP205 Düz Abutment Ø 3.8 mm H:5 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
2	SLM NP205 Düz Abutment Ø 3.8 mm H:5 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6
3	SLV NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
4	SLM NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6
5	SLV NP303 Düz Anatomik Abutment Ø 3.8 mm H:3 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
6	SLM NP303 Anatomik Düz Abutment Ø 3.8 mm H:3 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6
7	SLV NP362 Anatomik Açılı Abutment 25° Ø 3.8 mm H:3 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
8	SLM NP362 Anatomik Açılı Abutment 25° Ø 3.8 mm H:3 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6
9	SLV 30505 Multi Unit Abutment 0° Hex Ø 5.0 mm H:5 mm for Ø3.2 /3.7/4.3/5.0implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
10	SLM 30505 Multi Unit Abutment 0° Hex Ø 5.0 mm H:5mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6
11	SLV 30516 Multi Unit Açılı Abutment 30° Ø 5.0 mm H:4.5 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
12	SLM 35516 Multi Unit Açılı Abutment 30° Ø 5.0 mm H:4.5 mm for Ø3.7/4.2/4.7/5.2 implant için	SLM 3716- Medisolaris Merkür Hex İmplant Ø3.7 x 16.0 mm Platform Ø3.5 mm	SLM 18000 Bağlantı Vidası /Combining Screw M1.8
13	SLV 30605 Topuz Abutment Ø 4,5 mm H:5 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
14	SLM 30605 Topuz Abutment Ø 4,5 mm H:5 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6
15	SLV 30803 Tibase cad-cam Abutment H:3 mm Ø 3.8 mm	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6
16	SLM 30804 Tibase cad-cam Abutment H:3 mm Ø 3.8 mm	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6

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2. TEST SUMMARY

Client	Medisolaris Sağlık Hizmetleri San. ve Tic. Ltd. Şti.
Client's Adress	Fatih Mah. 1191 Sk. No:28/1 B Blok Gaziemir/İZMİR
Test Method	ISO/TS 13498: Dentistry - Torsion test of implant body/connecting part joints of endosseous dental implant systems
Test Name	Torsion Test For Endosseous Dental Implants
Specimens	Dental Implants(Mentioned at Table 1,2)

3. METHODOLOGY

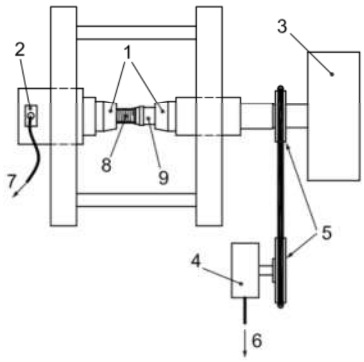

Figure 1. Test Configuration[6.1]

The tested dental implants 3D cad models are represented in Figure 2. Test set-up is shown in Figure 1.

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3.1. MATERIAL INFORMATION

The properties related to the material used in the structural analysis were used in the analysis as given below.

Table 3. Material&Mechanical Properties

	Elastic Modulus (MPa)	Poison Ratio	Density (g / cm3)
Ti6Al4V ELI - Grade 23*	113.800	0,342	4.43

*[7.2], [7.4]

3.2. GEOMETRY OF TEST MODELS

Figure 2.1	Figure 2.2
Figure 2.3	Figure 2.4

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Figure 2.5	Figure 2.6
Figure 2.7	Figure 2.8
Figure 2.9	Figure 2.10

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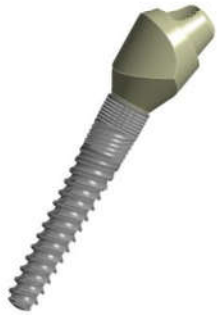	
Figure 2.11	Figure 2.12
Figure 2.13	Figure 2.14
Figure 2.15	Figure 2.16

Figure 2. Test Models

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3.3. LOADS AND SUPOORTS

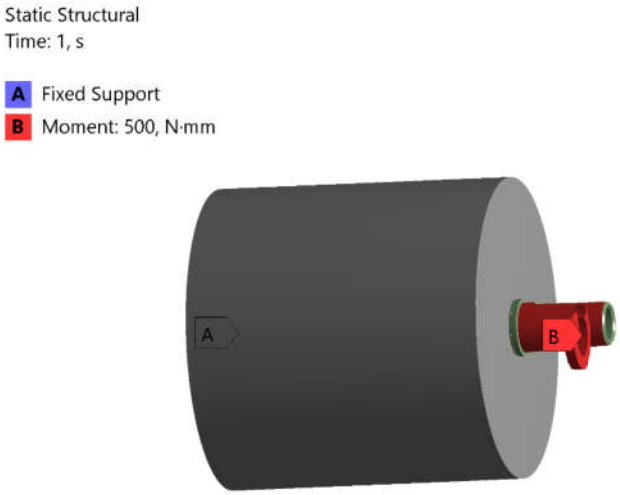

Figure 3. Loads and Supports (represent all models)

3.4. MESH MODEL OF DENTAL IMPLANTS

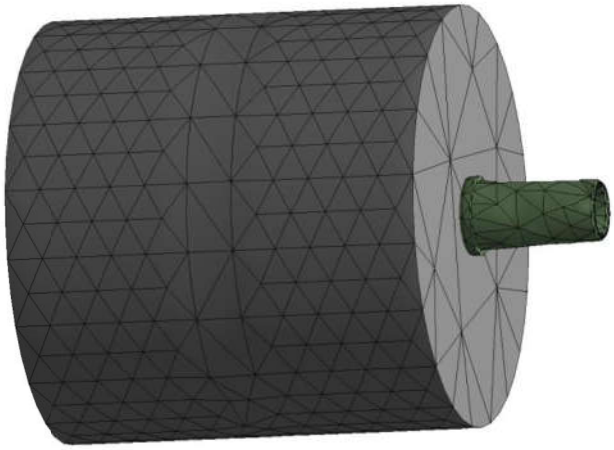

Figure 4. Mesh Model of Dental Implants (represent all models)

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4. TEST RESULTS

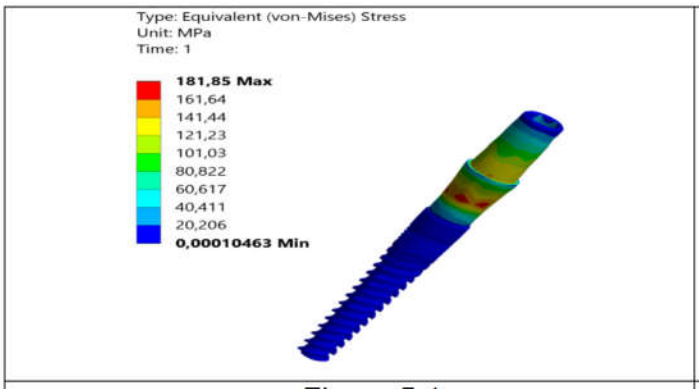

Figure 5.1

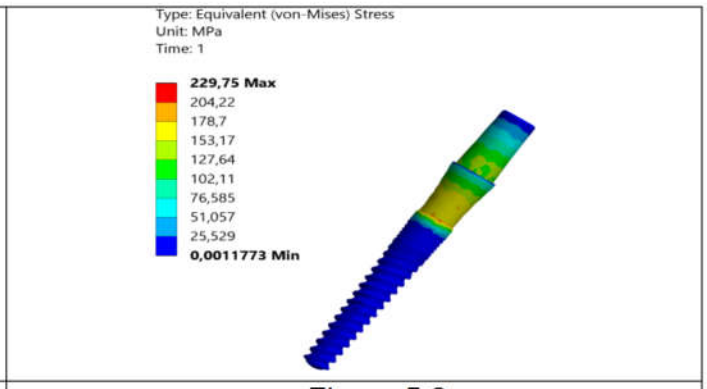

Figure 5.2

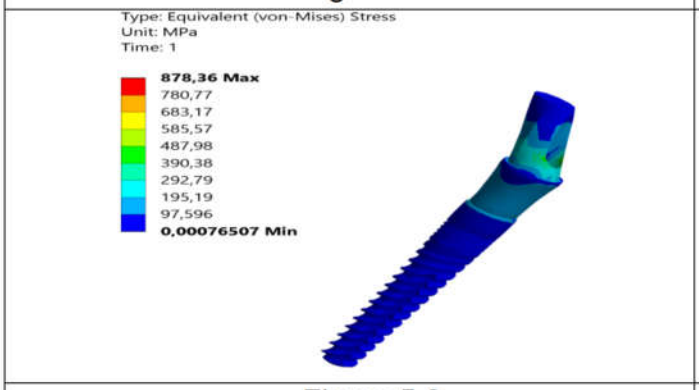

Figure 5.3

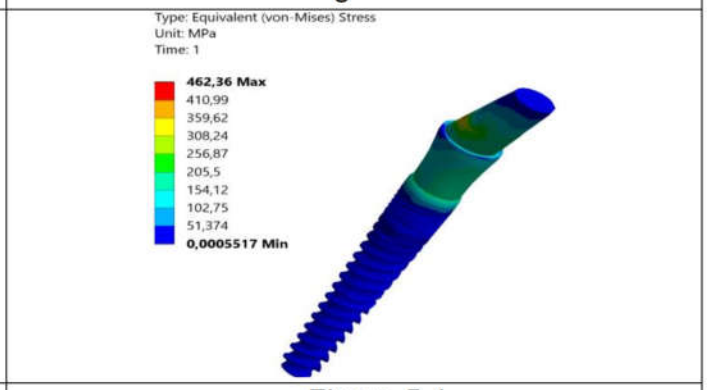

Figure 5.4

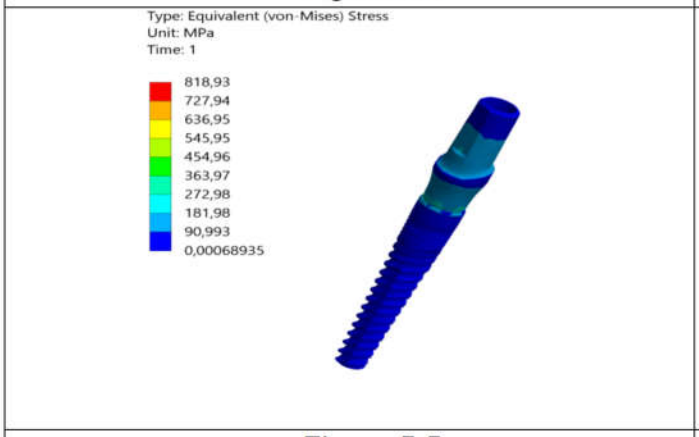

Figure 5.5

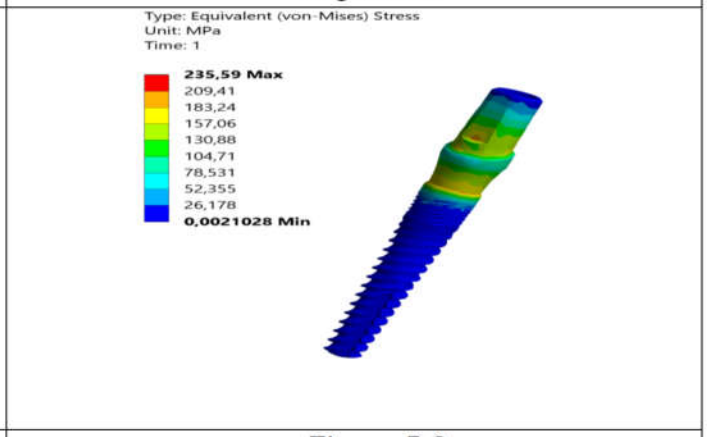

Figure 5.6

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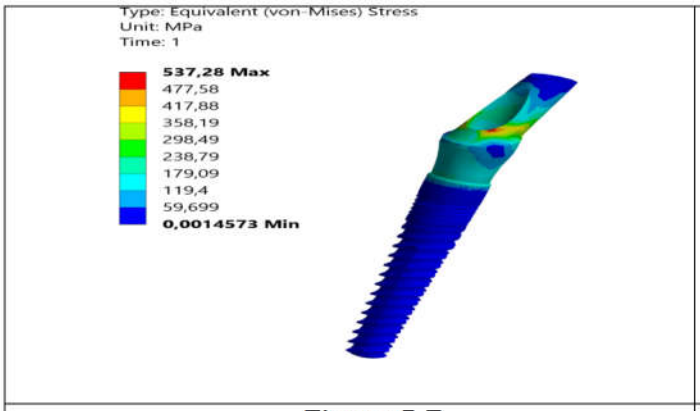

Figure 5.7

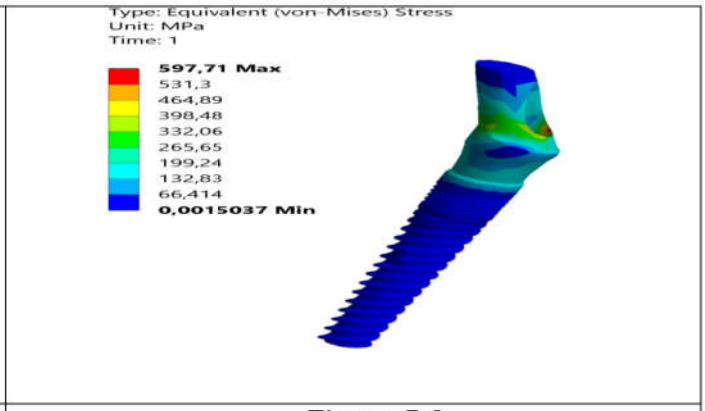

Figure 5.8

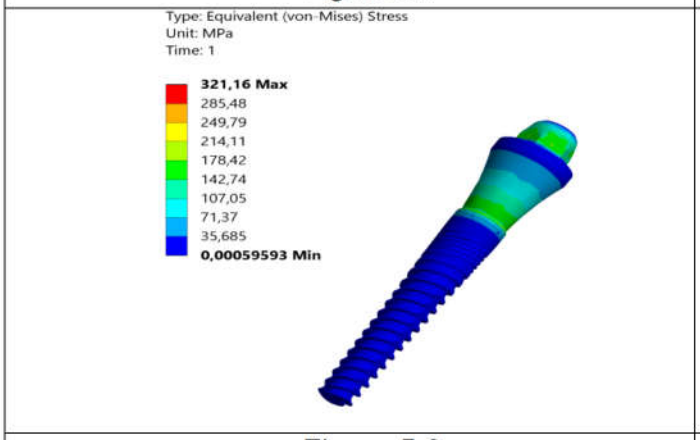

Figure 5.9

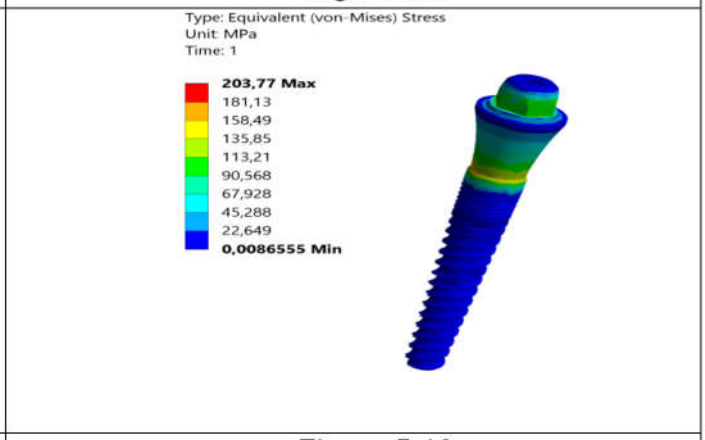

Figure 5.10

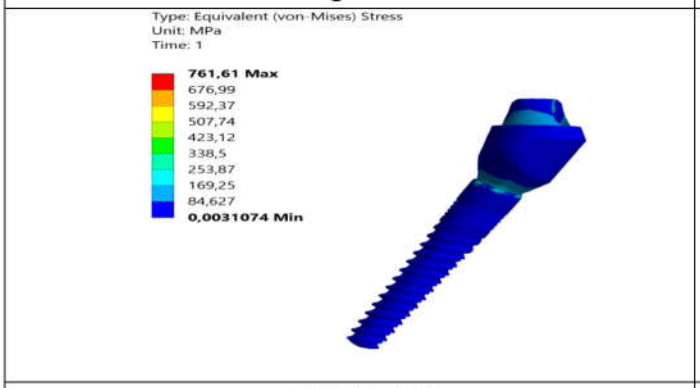

Figure 5.11

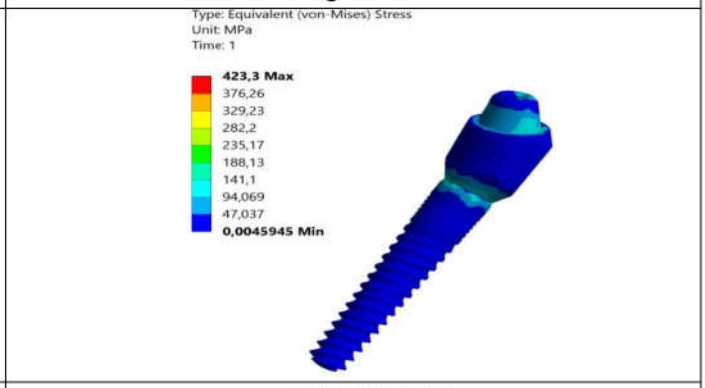

Figure 5.12

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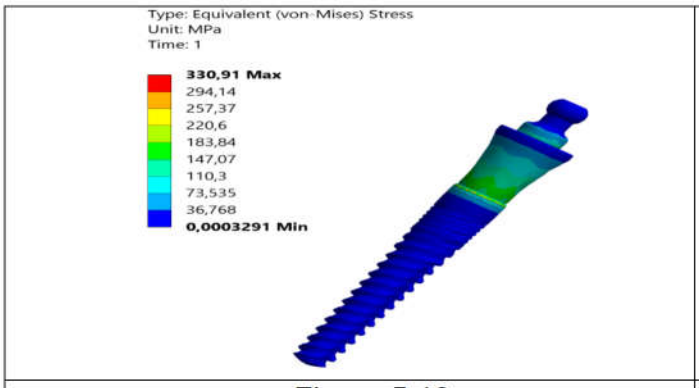

Figure 5.13

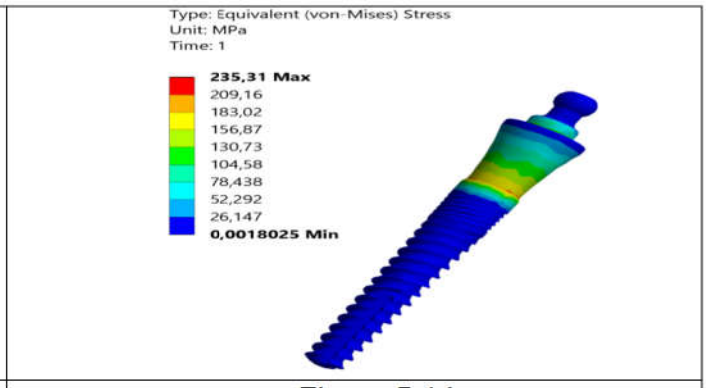

Figure 5.14

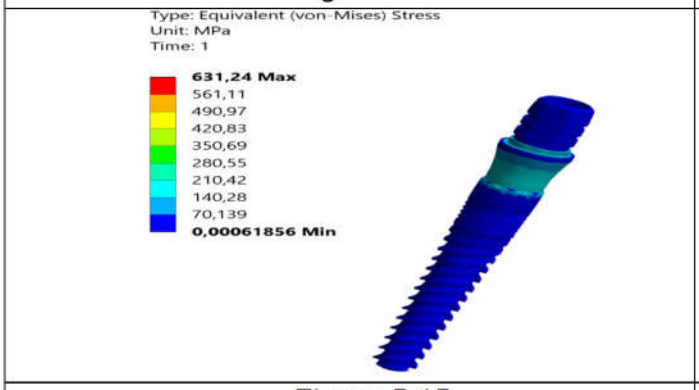

Figure 5.15

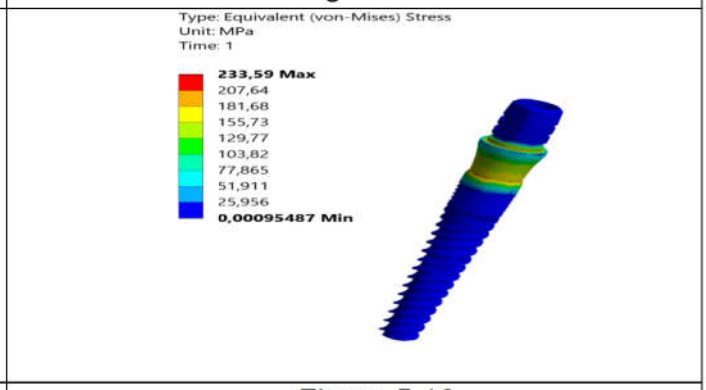

Figure 5.16

Figure 5. Test Results (MPa)

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5. CONCLUSION

Table 4. Test Results (Max. Stress (MPa))

NO	ABUTMENT	IMPLANT	SCREW	Maximum Equivalent Stress [MPa]
1	SLV NP205 Düz Abutment Ø 3.8 mm H:5 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	181,85
2	SLM NP205 Düz Abutment Ø 3.8 mm H:5 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	229,75
3	SLV NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	878,36
4	SLM NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	462,36
5	SLV NP303 Düz Anatomik Abutment Ø 3.8 mm H:3 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	818,93
6	SLM NP303 Anatomik Düz Abutment Ø 3.8 mm H:3 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	235,59
7	SLV NP362 Anatomik Açılı Abutment 25° Ø 3.8 mm H:3 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	537,28
8	SLM NP362 Anatomik Açılı Abutment 25° Ø 3.8 mm H:3 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	597,11
9	SLV 30505 Multi Unit Abutment 0° Hex Ø 5.0 mm H:5 mm for Ø3.2 /3.7/4.3/5.0implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	321,16
10	SLM 30505 Multi Unit Abutment 0° Hex Ø 5.0 mm H:5mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	203,77
11	SLV 30516 Multi Unit Açılı Abutment 30° Ø 5.0 mm H:4.5 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	761,61
12	SLM 35516 Multi Unit Açılı Abutment 30° Ø 5.0 mm H:4.5 mm for Ø3.7/4.2/4.7/5.2 implant için	SLM 3716- Medisolaris Merkür Hex İmplant Ø3.7 x 16.0 mm Platform Ø3.5 mm	SLM 18000 Bağlantı Vidası /Combining Screw M1.8	423,3
13	SLV 30605 Topuz Abutment Ø 4,5 mm H:5 mm for Ø3.2 /3.7/4.3/5.0 implant için	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	330,91
14	SLM 30605 Topuz Abutment Ø 4,5 mm H:5 mm for Ø3.2 implant için	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	235,31
15	SLV 30803 Tibase cad-cam Abutment H:3 mm Ø 3.8 mm	SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLV 16000 Bağlantı Vidası /Combining Screw M1.6	631,24
16	SLM 30804 Tibase cad-cam Abutment H:3 mm Ø 3.8 mm	SLM 3216-Medisolaris Neptün İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm	SLM 16000 Bağlantı Vidası /Combining Screw M1.6	233,59

According to the Finite Element Method test results, the dental implant specified in Number 3 ["SLV NP266 Açılı Abutment 25° Ø 3.8 mm H:4 mm for Ø3.2 /3.7/4.3/5.0 implant için, SLV 3216-Medisolaris Venüs Con İmplant Ø3.2 x 16.0 mm Platform Ø3.0 mm, SLV 16000 Bağlantı Vidası /Combining Screw M1.6"] was identified as the worst case.

[Note: This test report shall not be reproduced except in full without the written approval of the testing laboratory! The test results relate only to the items tested!]

6. TEST SUMMARY REPORT APPROVAL

The undersigned acknowledge they have reviewed the BIOMECHANICAL TEST REPORT and agree with the approach it presents. Changes to this **Test Report** will be coordinated with and approved by the undersigned or their designated representatives.

Marla Muhendislik San. ve Tic. A.S.	
Date:	05/01/2021
STAMP & SIGNATURE:	

7. REFERENCES

7.1. ISO/TS 13498: Dentistry - Torsion test of implant body/connecting part joints of endosseous dental implant systems

7.2. https://www.azom.com/article.aspx?ArticleID=9365

7.3. Medisolaris Product List

7.4. ASTM F136 - 13 - Standard Specification for Wrought Titanium-6Aluminum-4Vanadium ELI (Extra Low Interstitial) Alloy for Surgical Implant Applications (UNS R56401)

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